HW4 homework debrief.

Supp III

Consider that $\det A$ is a polynomial in the entries of A. In particular, it is a polynomial in x. In fact, since there is only one ~~term~~ entry that contains an x (and to the first power), we will see that $\det A$ has the form $f(x)=ax+b$; and so $f'(x)=a$.

We can cofactor expand iteratively, beginning in the first column:

$$\det A = -6(\text{constant}) - 4(\text{constant}) + 4(\text{constant}) - x\begin{vmatrix} 9 & -2 & 1 & 3 \\ 0 & 7 & 1 & -5 \\ 0 & 0 & -8 & 8 \\ 0 & 0 & 0 & -2 \end{vmatrix} + 4(\text{constant}).$$

And since this is upper triangular, we have

$$f(x) = \det A = -x\left(9\cdot 7\cdot(-8)(-2)\right) + \text{constant}$$

and $f'(x) = -1008.$ ∎

Supp IV

When confronted with a large computation, it is often a good idea to try a smaller similar calculation first. Let's do 4×4

$$C = \begin{pmatrix} 1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1 \end{pmatrix} \qquad I = \begin{pmatrix} 1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1 \end{pmatrix} \qquad D = \begin{pmatrix} 1&0&0&0\\0&2&0&0\\0&0&3&0\\0&0&0&4 \end{pmatrix}$$

$\det(C)=0$, since it has linearly dependent columns.

$\det(D) = 4! = 24$, since it is a diagonal matrix

$$\det(C-I) = \det\begin{pmatrix} 0&1&1&1\\1&0&1&1\\1&1&0&1\\1&1&1&0 \end{pmatrix}$$

We can create some symmetry by adding each of the other columns to the first column; and this does not affect the determinant:

$$= \det\begin{pmatrix} 3&1&1&1\\3&0&1&1\\3&1&0&1\\3&1&1&0 \end{pmatrix} = 3\det\begin{pmatrix} 1&1&1&1\\1&0&1&1\\1&1&0&1\\1&1&1&0 \end{pmatrix}$$

and now subtracting the first column off from each of the other three columns (which again does not affect the determinant):

$$= 3\det\begin{pmatrix} 1&0&0&0\\1&-1&0&0\\1&0&-1&0\\1&0&0&-1 \end{pmatrix} = 3\cdot(-1)^3 = -3.$$

$$\det(C - I + D) = \det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 4 \end{pmatrix}$$

This time, after subtracting the first column off from each of the other columns, we have

$$= \det\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{pmatrix} = 1 \cdot 3! = 6.$$

It is now not hard to generalize these calculations to the $n \times n$ case. Note that one should always check for special cases that work out a little differently!

$\det(C_n) = 0$ for $n \geq 2$ and $\det(C_1) = 1$

$\det(D_n) = n!$

$\det(C_n - I_n) = (n-1) \cdot (-1)^{n-1}$

$\det(C_n - I_n + D_n) = (n-1)!$

So for our original 100×100 problem, we have answers:

$\det(C) = 0$, $\det(D) = 100!$, $\det(C - I) = -99$

and $\det(C - I + D) = 99!$